Problem Solving with Areas

Exam code: 0607

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Updated on 28 January 2026

Problem-solving with areas

What is problem-solving?

- **Problem-solving**, usually has two key features:
  - A question is given as a **real-life scenario**
    - eg. Mary is painting a bedroom in her house...
  - There is usually **more than one topic of maths** you will need in order to answer the question
    - eg. Area and percentages

What are common problems that involve area?

- **Area** is a commonly used topic of **'real-world'** maths
  - For example, laying a carpet, painting a house or designing a sports field all involve area
- Typically, the 'real-world' scenarios also have a **cost**
  - A lot of area problems also involve **calculations with money**

How do I solve problems that involve area?

- There is often a **lot of text** in a problem-solving question, which can make it seem harder than it is
  - Avoid focusing **only** on what the question asks you, think about what you can do with the **information given**
    - This may lead you to think of **something else** you can do
    - Eventually you may be able to see your way to answering the **original question**
  - Think about the context of the question, **what makes sense?**
- Look out for **key information** in the text:
  - **Real-life context**
    - A question involving the **size** of a field, will mean looking at its **area**
  - **Key words**
    - Types of **measure**: area, length, cost, ...
    - **Conditions**: minimum, maximum, greatest, ...

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

need to do

- **Annotate diagrams** with information that you can work out
  - Remember to do this in pencil in case you need to erase it!

Examiner Tips and Tricks

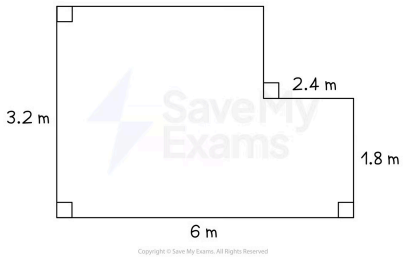
Even if you never get to a final answer always try to do some maths with the information from the question.

You are likely to score some extra marks from your working!

Worked Example

John wants a new carpet for the lounge in his house.

A sketch of his lounge is given below.



He gets quote from two local companies, Company A and Company B.

The amount they charge for laying a carpet is given below.

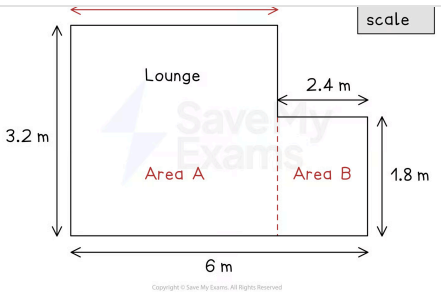
- Company A: Fixed price of £5.50 per square metre
- Company B: £6 per square metre for the first ten square metres, then £4 per square metre for anything over that.

Which company should John choose in order to keep the cost of laying the carpet to a minimum?

Answer:

Although this question doesn't specifically tell you you need to find the area, it is implied as the costs both use 'square metre'

The shape of the lounge is a compound shape consisting of two rectangles
Split the area into these two rectangles and find the missing distances by subtracting the smaller length (2.4 m) from the longer one (6 m)



Find the area of the lounge by adding the two areas together

$$\begin{aligned}\text{Total Area} &= \text{Area A} + \text{Area B}\\ &= (3.2 \times 3.6) + (2.4 \times 1.8)\\ &= 11.52 + 4.32\\ &= 15.84\ \text{m}^2\end{aligned}$$

Find the cost for each of the two companies separately

Company A:

$$\begin{aligned}\text{Total Cost} &= 15.84 \times 5.50\\ &= £87.12\end{aligned}$$

Company B:

$$\begin{aligned}\text{Total Cost} &= £6 \times \text{first } 10\ \text{m}^2 + £4 \times \text{remaining}\\ &= (6 \times 10) + 4 \times (15.84 - 10)\\ &= 60 + 23.36\\ &= £83.36\end{aligned}$$

John should choose Company B as it will cost £3.76 less than Company A

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